DO NOW - RECALL

Signed integers and binary arithmetic

Answer all four retrieval questions from previous lessons. Use black pen.

Name: _____ Class: _____ Date: _____

1. Choose a type

State the most suitable data type for a count of passengers and for whether a door is open.

2. Keep leading zeros

A room identifier is "0037". Choose its data type and justify why integer is unsuitable.

3. Read an unsigned byte

Convert 10100110 (base 2) to denary. Show your place-value sum.

4. Can it fit?

Write 200 in unsigned eight-bit binary. Explain why 300 will not fit in eight bits.

DO NOW - CHECK / OBJECTIVES

Correct and improve

Complete page 1 before looking at these model answers. Tick correct work in green; correct mistakes fully.

1. CHOOSE A TYPE

Integer for the passenger count; Boolean for whether a door is open.

2. KEEP LEADING ZEROS

String. An integer would remove the leading zeros, changing the identifier.

3. READ AN UNSIGNED BYTE

128 + 32 + 4 + 2 = 166.

4. CAN IT FIT?

200 = 11001000. An eight-bit unsigned integer ranges from 0 to 255, so 300 cannot fit.

Today: you should be able to...

- Represent positive and negative integers using sign magnitude and two's complement.
- Add and subtract fixed-width signed binary integers.
- Identify overflow and justify the range of a signed representation.

Key vocabulary

Sign magnitude — Sign bit followed by the magnitude.
Two's complement — Signed representation with a negative most significant place value.
Overflow — A mathematical result lies outside the representable range.
Negation — Changing a number to its additive inverse.

GREEN PEN - Improve one response

Choose a Do Now answer you corrected. Explain the exact misconception and write a stronger answer below.

NEW LEARNING / I DO

Read and annotate

Read the explanations, underline important terminology and add your own examples.

Two signed representations

In eight-bit sign magnitude, the leftmost bit indicates the sign: 0 positive; 1 negative. The other seven bits show magnitude. For example, 10000101 represents -5. Sign magnitude has two representations of zero.

Two's-complement place values

Eight-bit two's complement has weights -128, 64, 32, 16, 8, 4, 2, 1. The pattern 11111011 gives -128 + 64 + 32 + 16 + 8 + 2 + 1 = -5. The range is -128 to +127.

Negation and subtraction

To negate an eight-bit number, invert every bit and add one. To calculate A - B, add A to the two's complement of B, keeping the width fixed.

Carry is not the same as signed overflow

For signed eight-bit addition, an out-of-range mathematical result causes signed overflow. Two positive values producing a negative stored result (or vice versa) indicates overflow.

Always state whether a bit pattern is unsigned, sign magnitude or two's complement before interpreting it.

Explain in your own words

Compare 10000101 as an eight-bit sign-magnitude value and as an eight-bit two's-complement value. Explain why the values differ.

I DO - WORKED EXAMPLE

Represent -13 in two ways

Follow the complete worked example. Each step explains the reasoning, not just the result.

STEP 1 · START WITH POSITIVE 13

`00001101`

13 = 8 + 4 + 1. Write the fixed-width positive value first.

STEP 2 · SIGN MAGNITUDE

`10001101`

Change the sign bit to 1; preserve the seven-bit magnitude.

STEP 3 · INVERT BITS

`11110010`

Invert every bit of the positive eight-bit pattern.

STEP 4 · ADD ONE

`11110011`

The two's-complement representation of -13 is 11110011.

Check: -128 + 64 + 32 + 16 + 2 + 1 = -13. Both representations encode the same value using different rules.

Now repeat the method independently

Repeat the representation method for -21 in both eight-bit sign magnitude and eight-bit two's complement. Show the check.

WE DO - GUIDED PRACTICE

Try, compare, improve

Attempt each task. Discuss the method before checking with your teacher; show working, not only final answers.

1. Interpret a negative value

Convert 11100110 from eight-bit two's complement to denary. Use signed place values.

2. Add signed binary numbers

Calculate 00010111 + 00010011. Give the eight-bit result and explain whether signed overflow occurs.

3. Subtract using negation

Calculate 12 - 19 using eight-bit two's complement. Show the negation of 19 and the addition.

YOU DO - EXAM PRACTICE

Independent questions · Part 1

Use examination-style working. Do not look at model answers while you attempt these tasks.

1. Two representations [4 marks]

Represent -37 in (a) sign magnitude [2] and (b) two's complement [2]. Show the positive binary value and the method for each.

2. Interpret the same pattern [6 marks]

Determine the denary value of 10010110 interpreted as (a) unsigned [2], (b) sign magnitude [2] and (c) two's complement [2]. Show working.

YOU DO - EXAM PRACTICE

Independent questions · Part 2

Read the command words carefully. Where asked to explain, give the reason and use the correct technical language.

3. Signed addition [4 marks]

Add 00101101 and 00011011 as two's-complement integers. Show working, stored result, denary value and whether overflow occurs.

4. Subtraction [5 marks]

Calculate 18 - 29 using eight-bit two's complement. Show +29, -29, the addition and signed denary result.

5. Explain overflow [5 marks]

01110000 + 00100000 are added as eight-bit two's-complement values. State their denary values and stored result. Explain overflow and give the minimum signed bit width needed for the correct total.

EXIT TICKET / REFLECTION

Check your understanding

Write T or F and justify any false statement. Show your teacher your answers before checking.

1. Range

In eight-bit two's complement the signed range is -128 to +127.

2. Carry and overflow

A carry out of the leftmost bit always means signed overflow.

3. Negation

Two's complement negation in a fixed width uses invert-all-bits, then add one.

Self-assessment

Circle one: Confident / Nearly there / Need more help

GREEN PEN - Correct and explain

Choose one error from your work. Rewrite the correct method and explain why your first answer was incorrect.